VentureMiner AI

CI/CD

Document: 08-Engineering / 24_cicd
Version: v1.0
Date: 2026-07-20
Author: VentureMiner AI Documentation Team
Status: Approved
Project: VentureMiner AI — AI Venture Intelligence Platform

Table of Contents

Document 24 — CI/CD

> *The continuous integration and delivery pipeline. How a commit becomes a running service, on every push, with the right gates.*

Table of Contents

1. Purpose & Scope

This document is the contract for the CI/CD pipeline. It governs how code is built, tested, scanned, and deployed from the moment a commit is pushed.

2. Principles

1. Fast feedback. PR pipeline < 10 min.
2. Green main is sacred. Main is always deployable.
3. Boring by default. Use the platform (GitHub Actions + Argo CD) and avoid custom logic.
4. Gates are explicit. No hidden approval; no skipped steps.
5. Reproducible. Same input → same output, byte-for-byte where possible.

3. Pipeline overview

```
PR opened → PR pipeline (lint, unit, integration, contract, SAST, build)
PR merged → main pipeline (deploy to staging)
Tag pushed → release pipeline (deploy to canary, soak, deploy to prod)
Hotfix → hotfix pipeline (expedited to prod)
```

1. Lint & format — fail on any error.
2. Type check — TypeScript / mypy.
3. Unit tests — must pass; coverage reported.
4. Integration tests — must pass.
5. Contract tests — must pass.
6. SAST — Snyk; critical fails the build.
7. Dependency scan — Snyk + Dependabot; critical fails.
8. Build — produces an OCI image with content-addressable tag.
9. SBOM — generated and signed.
10. Reviewer assignments — CODEOWNERS.
11. Comment — coverage, SBOM, plan, image digest posted on PR.

5. Main branch pipeline

1. Build and tag with git-.
2. Push image to ECR.
3. Apply Terraform to staging.
4. Deploy to staging.
5. Run E2E smoke.
6. Update OpenAPI spec artifact.

6. Release pipeline

Weekly cadence:

1. Tag vX.Y.Z on main.
2. Build release image.
3. Deploy to canary (10% of traffic).
4. Soak for 24h; SLO monitoring.
5. If green → full deploy to production.
6. If red → hold; investigate; promote or revert.
7. Publish release notes (auto-generated from PRs + manual edits).

7. Hotfix pipeline

For SEV-1 / SEV-2 fixes:

1. Branch from main.
2. Same gates as PR, but a hotfix label skips the merge queue.

8. Build & test stages

- Hermetic builds — pinned tool versions; no network during build (except for fetching declared dependencies).
- Caching — dependencies cached per service.
- Parallelization — tests run in parallel where possible.
- Retries — flaky tests retried up to 2 times; persistent flakes quarantined.

9. Quality gates

Lint / format	PR merge, release tag
Type check	PR merge, release tag
Unit tests	PR merge, release tag
Integration tests	PR merge, release tag
Contract tests	PR merge, release tag
SAST	PR merge, release tag
Dependency scan	PR merge, release tag
E2E (smoke)	Release tag
Performance smoke	Release tag
Security smoke	Release tag

A gate failure blocks the next stage; a release tag with a failing gate cannot be cut.

10. Secrets in CI

- OIDC between GitHub Actions and AWS — short-lived credentials.
- No long-lived secrets in CI; secrets pulled at job start from AWS Secrets Manager via OIDC.
- Logs are scrubbed for secret patterns; CI fails if a known secret is found.

11. Pipeline observability

- Datadog CI Visibility for build & test traces.
- GitHub checks for status.
- Slack notifications on tag, on hold, on release.
- Pipeline dashboards for DORA metrics (lead time, deploy freq, MTTR, change fail rate).

12. Pipeline cost

- Per-service budget for CI minutes.

13. Appendix

13.1 Revision history

v0.5	2026-07-20	Doc Team	All sections drafted
v1.0	2026-07-20	Doc Team	First approved version

13.2 Cross-references

- DevOps: Document 23.
- Testing: Document 22.
- Security: Document 21.

End of Document 24 — CI/CD. The pipeline is the spine of the engineering org.